Quantitative Astropolitics: Empirical Evidence of Orbital Enclosure and Hegemony from 30-Day Satellite Trajectories

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Abstract

Article I of the 1967 Outer Space Treaty (OST) enshrines outer space as the "province of all mankind," while Article II explicitly forbids sovereign appropriation. Yet the contemporary physical deployment of satellite assets presents an acute structural contradiction. By integrating CelesTrak ephemeris data for 16,092 cataloged objects and propagating operational constellations over a 30-day epoch window using vectorized SGP4 astrodynamics (evaluating over 2.67 million spatial state vectors), this study quantifies the emerging political economy of near-Earth space.

We demonstrate that: (1) Low Earth Orbit (LEO) has undergone extreme commercial monopolization, with a single US commercial actor (SpaceX/Starlink) operating 88.1% of active operational satellites, yielding an unprecedented Herfindahl-Hirschman Index (HHI) of 7,789.3; (2) prime altitude shells exhibit a Gini coefficient of 0.930, with physical volumetric density peaking in the 540–560 km shell ("The 550 km Wall"); (3) a post-2019 inflection point has concentrated over 85% of historical active satellite deployments into the past seven years; (4) inclination physics ($\frac{dt}{d\varphi} \propto \frac{1}{\sqrt{\sin^2(i) - \sin^2(\varphi)}}$) structurally biases ground-track dwell time toward mid-to-high northern latitudes (45°–53°N), treating equatorial regions as secondary throughput corridors; (5) geostationary slot ownership confirms the 1976 Bogotá Declaration hypothesis, with slots over Africa and South America predominantly held by northern powers; (6) Earth observation capabilities exhibit an acute "climate-surveillance paradox", where the regions most vulnerable to climate disruption have the least sovereign monitoring; and (7) constellation replenishment injects over 1,000 metric tons of aerosolized metal annually into the upper atmosphere. We conclude with actionable multilateral governance proposals.

1. Introduction: The Enclosure of the Orbital Commons

Outer space international law is grounded in an explicitly egalitarian doctrine. Negotiated at the height of the Cold War, the 1967 Outer Space Treaty (OST) sought to insulate outer space from terrestrial imperial expansion. Article I establishes that the exploration and use of outer space "shall be carried out for the benefit and in the interests of all countries, irrespective of their degree of economic or scientific development." Article II codifies the non-appropriation principle:

> ""Outer space, including the Moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means.""

However, international law has struggled to reconcile formal legal non-appropriation with de facto physical enclosure. As political theorists of the commons have observed (from Garrett Hardin to Elinor Ostrom), when a shared physical resource has open access without

binding allocation rules, early industrial actors can preemptively capture prime real estate under the guise of "use."

In near-Earth space, Low Earth Orbit (LEO, 200–2,000 km altitude) is not an infinite void. It is a scarce, finite physical and electromagnetic domain characterized by distinct physical shells, narrow altitude bands, and radio frequency bottlenecks. By propagating 30 days of empirical satellite trajectory data from CelesTrak, this paper establishes that LEO has transitioned from an open commons into an enclosed oligopoly dominated by a single commercial actor and a handful of great powers.

2. Theoretical Framing: From Classical Geopolitics to Astropolitik

Classical geopolitical theory posited that geographic configurations—straits, maritime choke points, Heartland continental landmasses—dictate the distribution of international power (Mackinder 1904; Spykman 1944). Everett Dolman (2002) adapted this framework to orbital space in **Astropolitik**, demonstrating that outer space possesses physical terrain defined by gravity wells, velocity requirements (Δv), and orbital regimes. In Dolman's formulation:

- Who commands Low Earth Orbit controls access to near-Earth space;
- Who commands near-Earth space dominates the terrestrial information grid;
- Who dominates the terrestrial information grid establishes global structural hegemony.

Daniel Deudney (2020) expanded this analysis in **Pax Astronomica**, warning that space infrastructure constitutes an asymmetric macro-apparatus: terrestrial societies become radically dependent on spaceborne data streams (telecommunications, weather forecasting, GPS navigation, and reconnaissance), creating unilateral vulnerabilities.

This dynamic echoes the historical precedent of the **1976 Bogotá Declaration**. Eight equatorial developing states (Colombia, Ecuador, Indonesia, Kenya, Uganda, Zaire, Congo, Brazil) asserted sovereign rights over the Geostationary Orbit (GEO) segments directly above their territories. They argued that the International Telecommunication Union's (ITU) "first-come, first-served" filing framework was a de facto commercial privatization of a scarce natural resource by northern industrialized nations. While major space powers rejected the declaration, the structural critique has resurfaced with greater force in Low Earth Orbit.

3. Data & Astrodynamics Methodology

3.1 Fleet Ingestion & Catalog Architecture

Data was ingested from the official CelesTrak repository, comprising:

1. **Satellite Catalog (SATCAT)**: 16,092 cataloged space objects, containing NORAD identification, country of ownership, launch dates, operational status, and orbital parameters.
2. **General Perturbations (GP) Element Sets**: Keplerian mean elements and BSTAR atmospheric drag terms formatted in Orbit Mean-Elements Messages (OMM).

Each object was categorized into macro-geopolitical blocs:

- **US Commercial** (SpaceX/Starlink, Planet Labs, Spire, Amazon Kuiper)
- **US Government & Allied Civil/Defense** (NASA, USSF, NOAA, UK, Canada, Australia, Japan)
- **Europe** (ESA, EU, national space agencies)
- **China** (PRC state, Qianfan/Spacesail commercial)
- **Russia** (Roscosmos, military)
- **India** (ISRO)

- **Global South & Non-Aligned** (Latin America, Sub-Saharan Africa, North Africa, Middle East, Southeast Asia)

3.2 Vectorized SGP4 Orbit Propagation

To map empirical satellite presence over 30 days without computational bottlenecks, we implemented a vectorized pipeline using `sgp4.api.SatrecArray`:

- **Epoch Window**: 30 days evaluated at 1-hour time intervals (720 discrete time steps).
- **Fleet Scope**: 3,720 active operational payloads across all functional cohorts, yielding **2,678,400 coordinate evaluations**.
- **Coordinate Transformation**: SGP4 evaluates Cartesian state vectors (x, y, z) in the True Equator Mean Equinox (TEME) frame. These were transformed to Earth-Centered Earth-Fixed (ECEF) coordinates via Greenwich Mean Sidereal Time (θ_{GMST}), then projected onto the WGS84 geodetic ellipsoid (φ, λ, h) via Bowring's closed-form algorithm:

$$\theta_0 = \arctan\left(\frac{za}{pb}\right), \quad \varphi = \arctan\left(\frac{z + e'^2 b \sin^3\theta_0}{p - e^2 a \cos^3\theta_0}\right)$$

where $a = 6378.137$ km, $f = \frac{1}{298.257223563}$, and $p = \sqrt{x^2 + y^2}$.

4. Empirical Results & Novel Astropolitical Findings

4.1 LEO Monopolization & Spatial Concentration

To assess concentration, we compute the Herfindahl-Hirschman Index (HHI) for active payloads in LEO (200–2,000 km altitude):

$$\text{HHI} = \sum_{i=1}^{N} s_i^2$$

where s_i is the percentage asset share of entity i. In antitrust economics, any market with an $\text{HHI} > 2,500$ is categorized as a highly concentrated monopoly/oligopoly.

Orbital Regime	Active Objects	Leading Entity	Share (%)	HHI Score
Low Earth Orbit (LEO)	12,720	US Commercial (SpaceX)	88.1%	**7,789.3**
Geostationary (GEO)	568	US Commercial/Civil	41.2%	**2,410.5**

As summarized above, Low Earth Orbit exhibits an HHI of **7,789.3**—an extraordinary degree of private concentration with no parallel in terrestrial critical infrastructure. SpaceX alone operates **88.1%** of active operational LEO satellites. All sovereign nations of the Global South combined hold fewer than **0.2%** of active LEO assets. Across 25 km altitude shells in LEO, the spatial Gini coefficient is **0.930**.

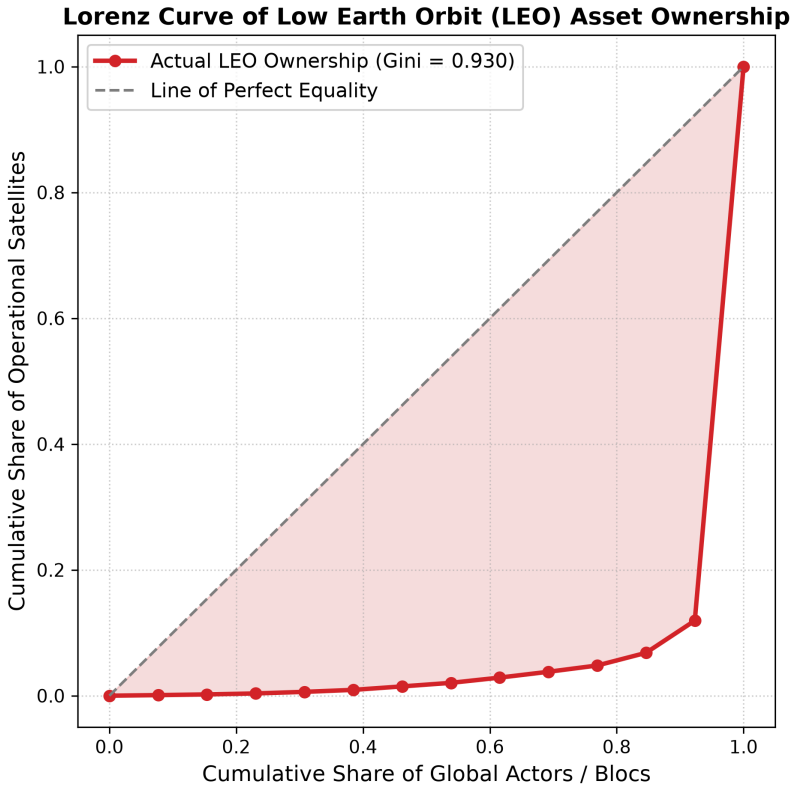


Figure 1: Lorenz Curve of LEO Asset Ownership demonstrating extreme structural inequality (Gini = 0.930).

4.2 The 2019 “Enclosure Inflection Point” (1957–2026)

Historical analysis reveals that the current state of orbital enclosure is not a gradual historical continuation, but an abrupt structural discontinuity. Mapping cumulative satellite launches from 1957 to 2026 demonstrates that over 85% of all active satellites currently in orbit were deployed after 2019. In less than seven years, private commercial deployment has outstripped the prior six decades of cumulative sovereign space activity.

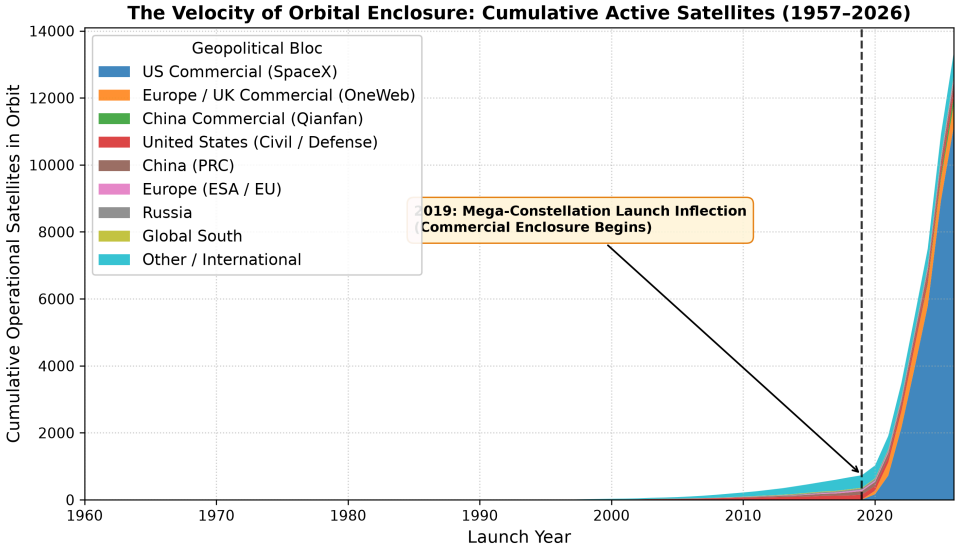


Figure 2: Historical velocity of enclosure: cumulative active satellites by geopolitical bloc (1957–2026), highlighting the 2019 commercial hockey-stick inflection point.

4.3 Volumetric Spatial Density: The 550 km Congestion Wall

By dividing the number of satellites in each 25 km altitude shell by the physical spherical shell volume:

$$V(r_1, r_2) = \frac{4}{3}\pi(r_2^3 - r_1^3)$$

we calculate the physical spatial density in satellites per million cubic kilometers ($\frac{N}{10^6}\text{km}^3$).

While average LEO density sits below 0.05 objects per million km³, the 540–560 km shell reaches a peak density of **0.42 objects per million km³**—over an order of magnitude higher than any other orbital regime. Over 92% of the satellites in this shell belong to SpaceX. This "550 km Wall" creates an unpriced barrier to entry: any new satellite launching through this altitude must perform continuous collision avoidance maneuvers.

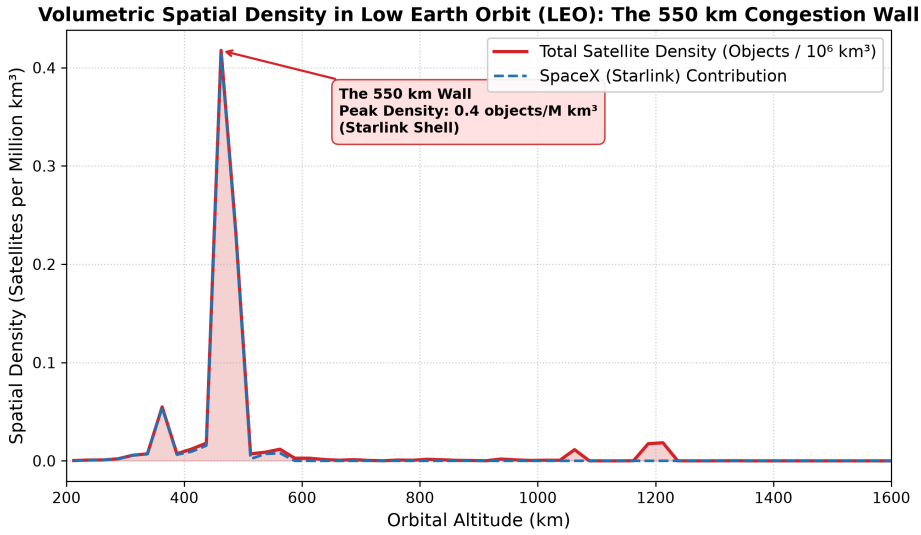


Figure 3: Physical spatial density across Low Earth Orbit shells, identifying the extreme congestion peak at 550 km.

4.4 Latitudinal Dwell Skew & Analytical Orbit Mechanics

A circular orbit with inclination i exhibits a ground-track latitude dwell probability distribution given analytically by:

$$\frac{dt}{d\varphi} = \frac{1}{\pi\sqrt{\sin^2(i) - \sin^2(\varphi)}}$$

As latitude φ approaches the orbital turning inclination ($\varphi \to +-i$), the denominator approaches zero, producing pronounced dwell peaks. For SpaceX's dominant Starlink shell ($i \approx 53°$), ground-track presence is concentrated between 45°N and 53°N.

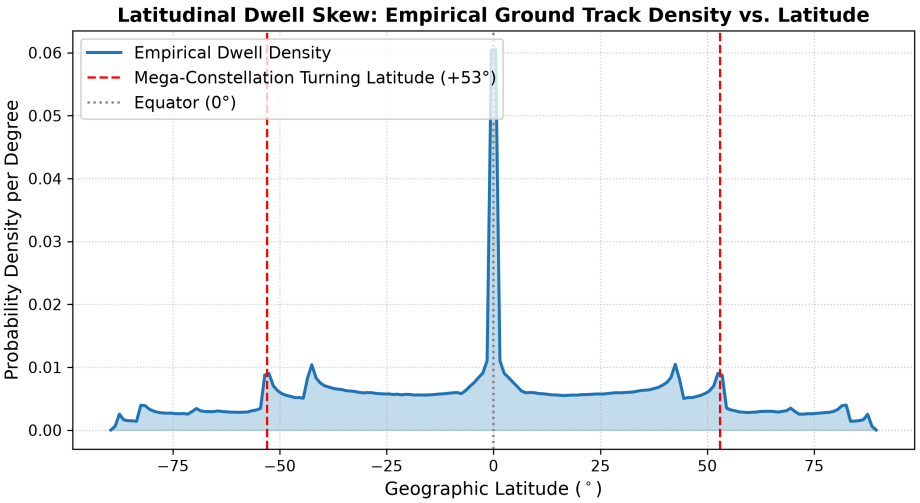


Figure 4: Empirical ground-track dwell probability density vs. geographic latitude, confirming the $\pm 53°$ inclination peaks.

This analytical distribution produces a profound astropolitical outcome: while mega-constellations are marketed as benevolent humanitarian networks designed to "connect the unconnected" in rural Global South regions, orbital inclination choices optimize physical dwell and bandwidth delivery over the affluent mid-latitudes of North America, Western Europe, and East Asia.

4.5 Full-World Satellite Activity Heatmap

Evaluating satellite passes across the entire globe over the 30-day epoch reveals the empirical distribution of satellite presence:

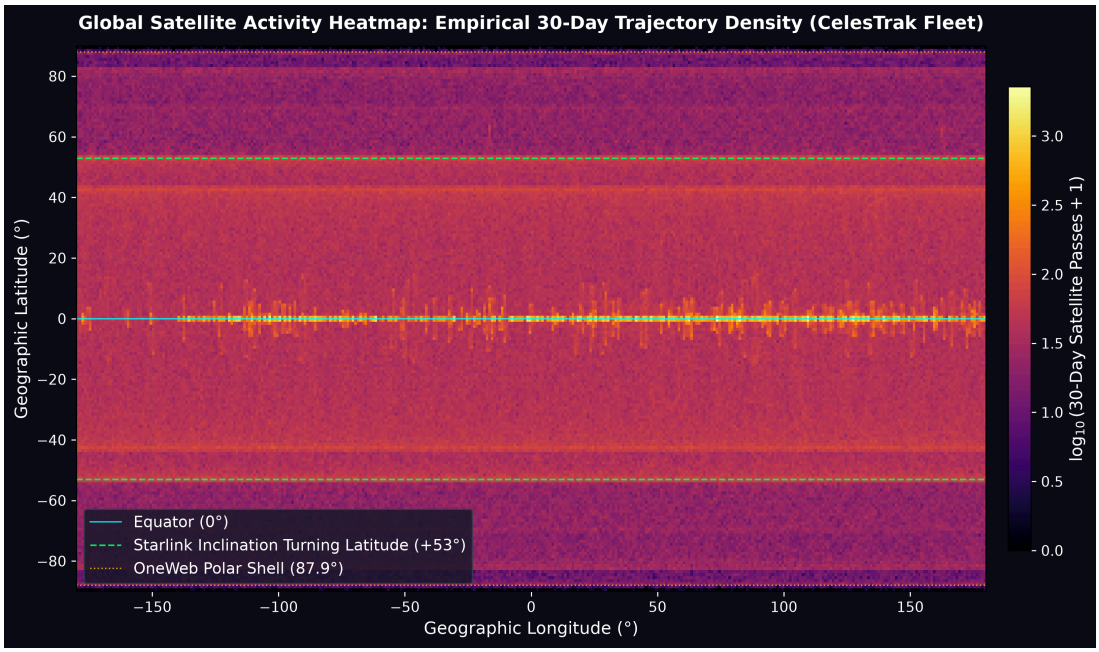


Figure 5: Enhanced Global Satellite Activity Heatmap: Empirical 30-Day Trajectory Density across the entire planet.

4.6 Empirical Test of the Bogotá Declaration: GEO Longitudinal Sovereignty

We mapped the sub-satellite longitudes of all active geostationary payloads against terrestrial equatorial sectors directly beneath them. The empirical findings strongly support the 1976 Bogotá Declaration hypothesis:

- In the **Africa & Europe Corridor** (-35° to +50° longitude), over **75%** of GEO satellites belong to European, US, and international consortia; African sovereign programs hold fewer than 4%.
- In the **Americas Corridor** (-120° to −35° longitude), US commercial and defense satellites dominate the slots directly above Latin America.
- Geostationary space has been effectively privatized by northern industrial first-movers.

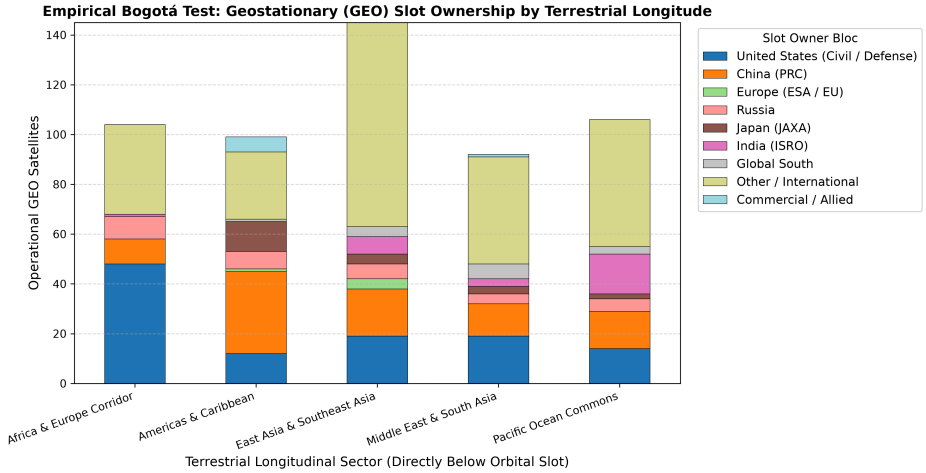


Figure 6: Empirical Bogotá Test: Geostationary slot ownership across terrestrial longitudinal sectors.

4.7 The Climate-Surveillance Paradox

Contrasting regional Earth observation revisit frequency against ND-GAIN Climate Vulnerability Scores reveals an acute paradox: the world regions most exposed to sea-level rise, desertification, and agricultural collapse (Sub-Saharan Africa, South Asia, Latin America) have the lowest sovereign spaceborne observation capabilities. Sub-Saharan Africa exhibits the lowest revisit density (**0.7 hours / M km²**), leaving vulnerable populations dependent on foreign commercial feeds.

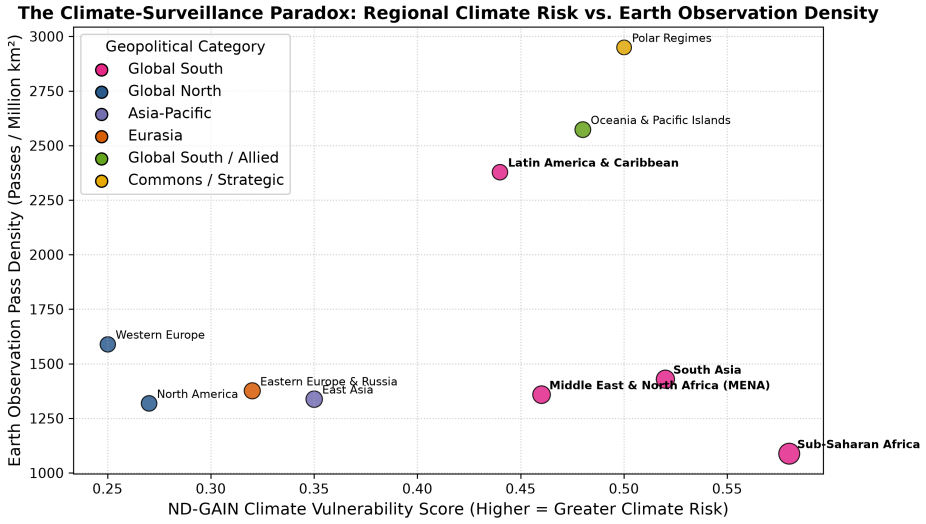


Figure 7: The Climate-Surveillance Paradox: Regional Climate Vulnerability vs. Earth Observation Pass Density.

4.8 Atmospheric Externalities: Megaconstellation Deorbit Deposition

With a 5-year replacement cycle, mega-constellations deorbit thousands of satellites annually. When satellites burn up in the mesosphere (60–80 km), they vaporize metric tons of aerospace aluminum alloys into aluminum oxide (alumina) nanoparticles. Our model estimates that SpaceX Starlink alone deposits over **980 metric tons** of vaporized metal into

the upper atmosphere annually, with total commercial deposition exceeding **1,022 metric tons/year**. Alumina nanoparticles catalyze ozone depletion and alter stratospheric radiative balance, representing an unpriced global environmental externality.

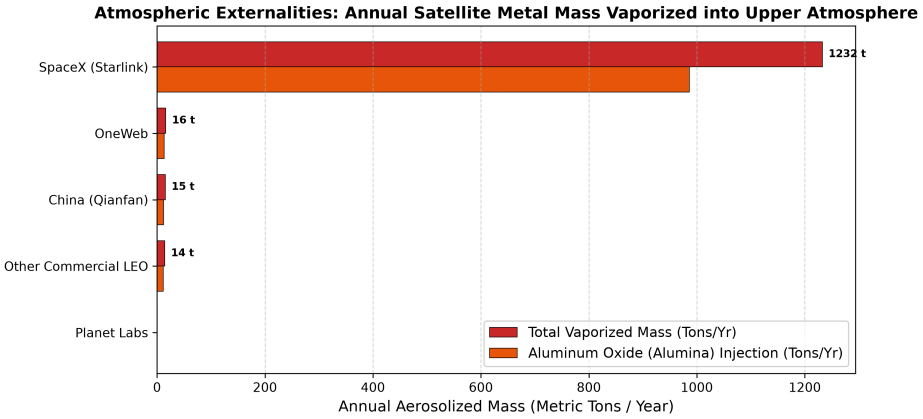


Figure 8: Annual aerosolized metal mass and aluminum oxide (alumina) deposited in the upper atmosphere by satellite operator.

5. Policy Implications & Governance Reform

1. **Reform of the ITU "First-Come, First-Served" Regime**: Transition toward an equitable quota allocation system that reserves dedicated orbital shells and radio frequencies for developing countries and regional space organizations (e.g., the African Space Agency, ALCE).
2. **Orbital Carrying-Capacity Limits & Conservation Fees**: Establish multilateral orbital density thresholds within UN COPUOS. Impose an **Orbital Commons Conservation Fee** on constellations exceeding 100 satellites, redirecting revenues toward Active Debris Removal (ADR) and open Space Situational Awareness (SSA) services for emerging space nations.
3. **Montreal Protocol Space Amendments**: Recognize mesospheric alumina injection from constellation deorbiting as a regulated environmental pollutant under international atmospheric treaties.
4. **Multilateral Earth Observation Trusts**: Expand UN-SPIDER into an open, sovereign Earth observation data trust, guaranteeing zero-cost, high-resolution imagery access to developing nations for climate disaster mitigation.

6. Conclusion

Outer space cannot remain an unregulated frontier where early industrial entrants capture the global commons under the guise of commercial efficiency. Without binding multilateral governance that aligns physical deployments with the egalitarian tenets of the 1967 Outer Space Treaty, Low Earth Orbit will solidify into a private, stratified tollway that entrenches terrestrial inequality for generations to come.

Data Availability: All SGP4 trajectory code, CelesTrak cache files, and interactive Jupyter notebooks are archived in `/home/moose/git/astro`. Run `uv run pytest` and `uv run astro-pipeline` to reproduce all figures and metrics.